OPEN SUBSPACES OF SCHEMES ARE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1897-1905, following the statement at 1890-1895.

Allowed standard prerequisites (editorial).

- Standard opens of affine schemes are affine. lemma-standard-open-affine (card L26419, cited at ~1903). D(f) is affine.

Source reference. Original lemma 01I3: sources/stacks-schemes/pinned-original-source.tex, source lines 1183–1183. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. Let $j : U \to X$ be an open immersion of locally ringed spaces. Then U is a scheme. In particular, any open subspace of X is a scheme.*

Proof. Let $U \subset X$. Let $u \in U$. Pick an affine open neighbourhood $u \in V \subset X$. Because standard opens of V form a basis of the topology on V we see that there exists a $f \in \mathcal{O}_V(V)$ such that $u \in D(f) \subset U$. And $D(f)$ is an affine scheme by Lemma 01I3. This proves that every point of U has an open neighbourhood which is affine. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.